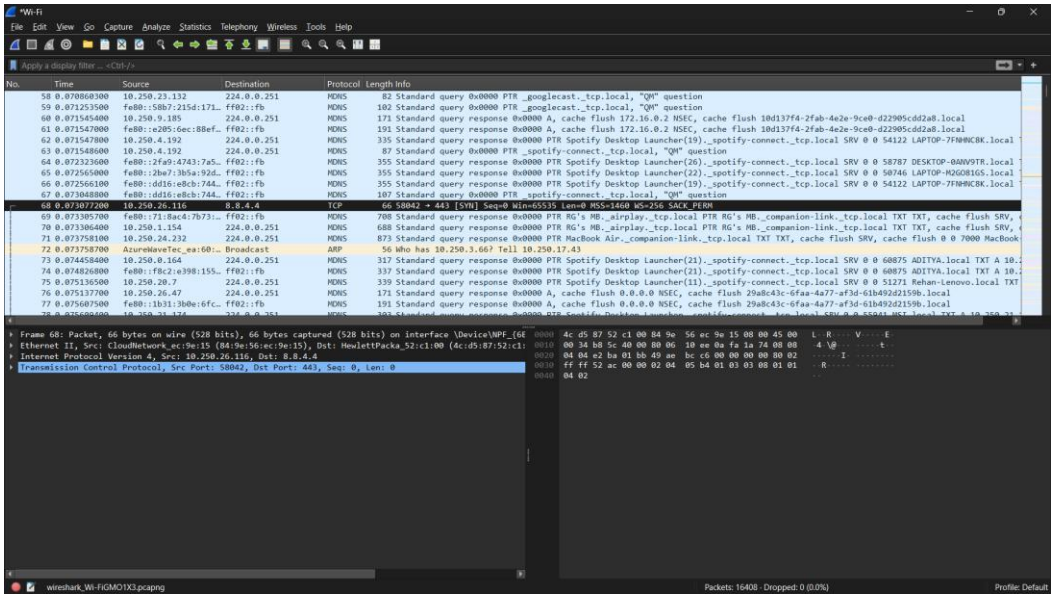
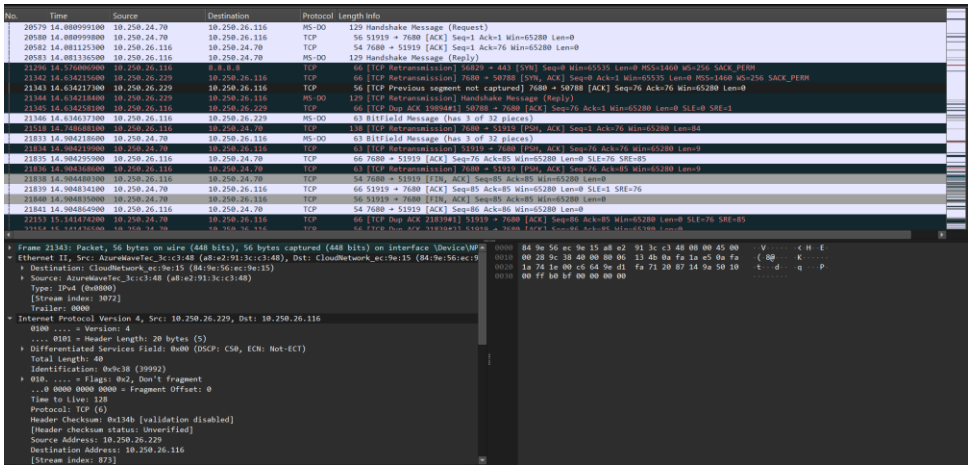
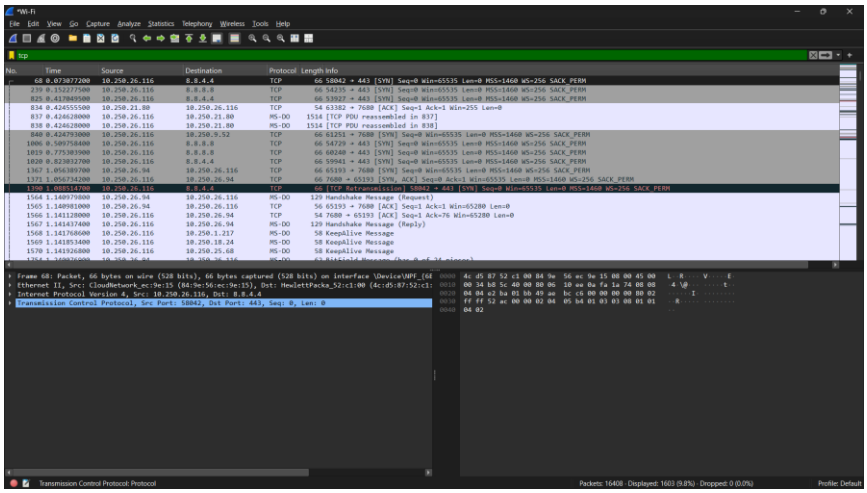
Skill– 1

Step 1 to 3

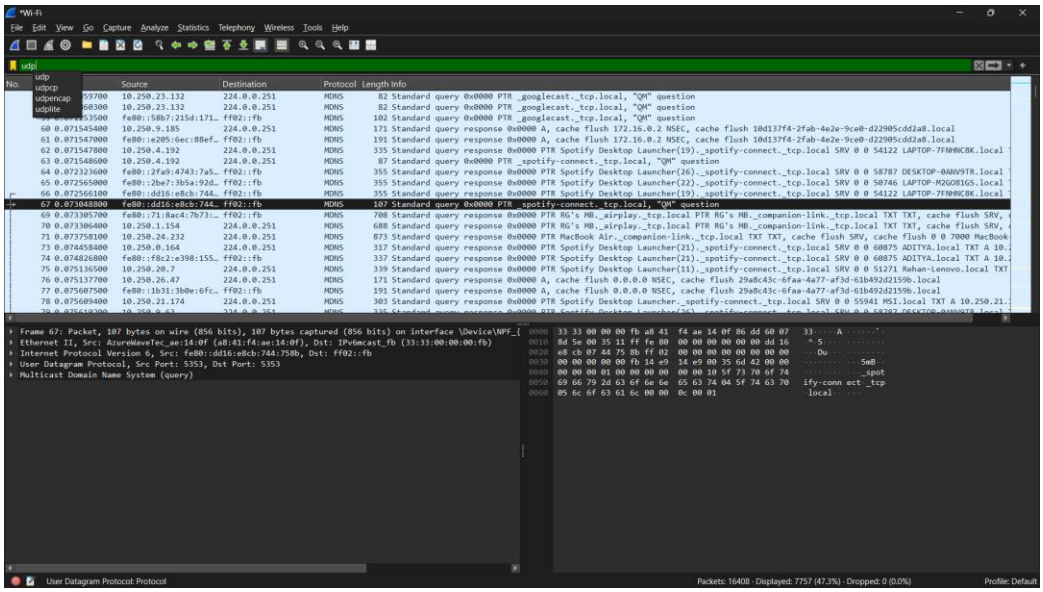


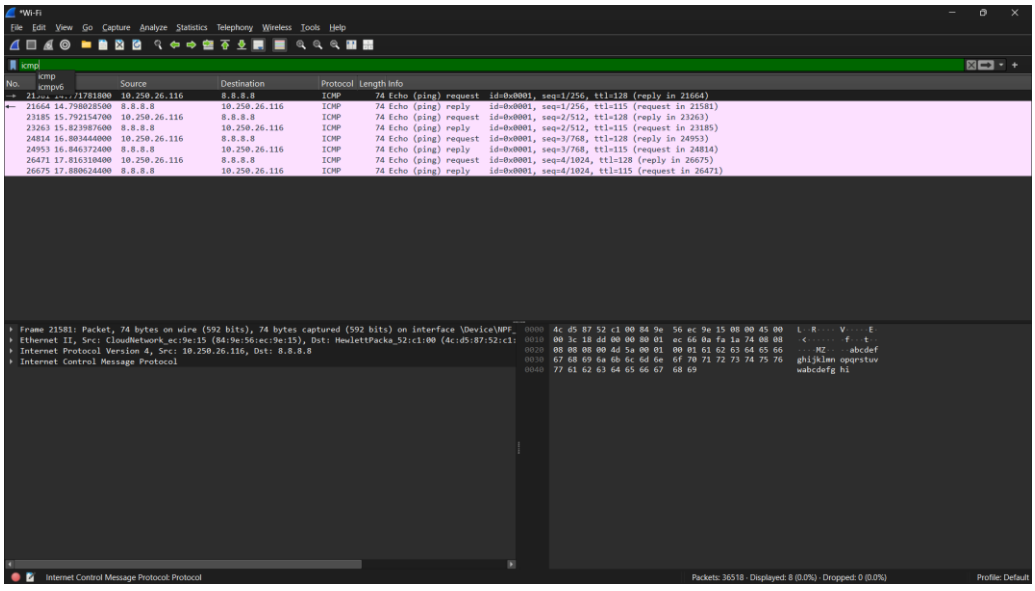
Step -4



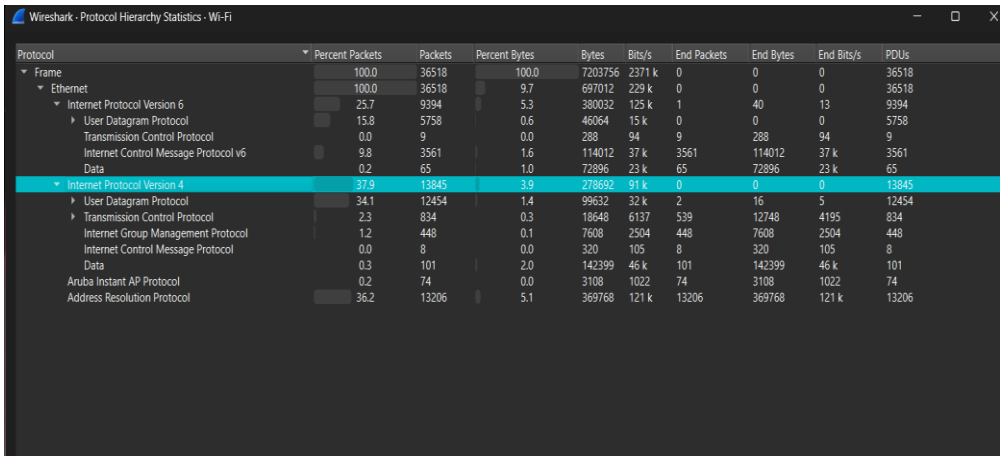


```
Frame 21343: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface \Device\NP
Ethernet II, Src: AzureWaveTec_3c:c3:48 (a8:e2:91:3c:c3:48), Dst: CloudNetwork_ec:9e:15 (84:9e:56:ec:9
   Destination: CloudNetwork_ec:9e:15 (84:9e:56:ec:9e:15)
   Source: AzureWaveTec_3c:c3:48 (a8:e2:91:3c:c3:48)
   Type: IPv4 (0x0800)
   [Stream index: 3072]
   Trailer: 0000
Internet Protocol Version 4, Src: 10.250.26.229, Dst: 10.250.26.116
   0100 .... = Version: 4
   .... 0101 = Header Length: 20 bytes (5)
   Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
   Total Length: 40
   Identification: 0x9c38 (39992)
   010. .... = Flags: 0x2, Don't fragment
   ...0 0000 0000 0000 = Fragment Offset: 0
   Time to Live: 128
   Protocol: TCP (6)
   Header Checksum: 0x134b [validation disabled]
   [Header checksum status: Unverified]
   Source Address: 10.250.26.229
   Destination Address: 10.250.26.116
   [Stream index: 873]
```



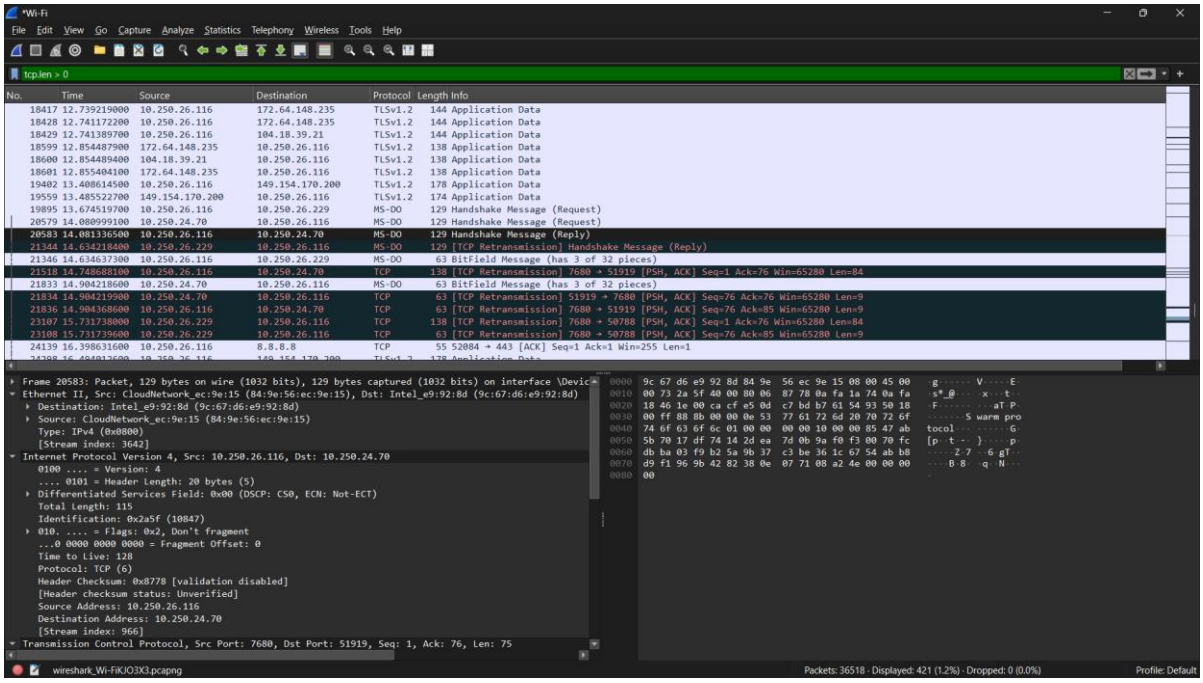


Step-4



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	36518	100.0	7203756	2371 k	0	0	0	36518
Ethernet	100.0	36518	9.7	697012	229 k	0	0	0	36518
Internet Protocol Version 6	25.7	9394	5.3	380032	125 k	1	40	13	9394
User Datagram Protocol	15.8	5758	0.6	46064	15 k	0	0	0	5758
Transmission Control Protocol	0.0	9	0.0	288	94	9	288	94	9
Internet Control Message Protocol v6	9.8	3561	1.6	114012	37 k	3561	114012	37 k	3561
Data	0.2	65	1.0	72896	23 k	65	72896	23 k	65
Internet Protocol Version 4	37.9	13845	3.9	278692	91 k	0	0	0	13845
User Datagram Protocol	34.1	12454	1.4	99632	32 k	2	16	5	12454
Transmission Control Protocol	2.3	834	0.3	18648	6137	539	12748	4195	834
Internet Group Management Protocol	1.2	448	0.1	7608	2504	448	7608	2504	448
Internet Control Message Protocol	0.0	8	0.0	320	105	8	320	105	8
Data	0.3	101	2.0	142399	46 k	101	142399	46 k	101
Aruba Instant AP Protocol	0.2	74	0.0	3108	1022	74	3108	1022	74
Address Resolution Protocol	36.2	13206	5.1	369768	121 k	13206	369768	121 k	13206

Step-5



Step – 6

TCP AND UDP HEADER OVERHEAD CALCULATIONS

Values taken from the selected Wireshark packets

1. TCP PACKET (Ethernet + IPv4 + TCP)

PACKET VALUES		CALCULATIONS
Frame length	**129 bytes**	Header overhead = 14 + 20 + 20 = 54 bytes
Ethernet header	**14 bytes**	Payload = 115 - 20 - 20 = 75 bytes
IPv4 header	**20 bytes**	Efficiency = (75 / 129) x 100
TCP header	**20 bytes**	Efficiency = 58.14%
IPv4 total length	**115 bytes**	
TCP segment / payload	**75 bytes**	

TCP RESULT: 54 bytes overhead | 75 bytes payload | 58.14% efficiency

2. UDP PACKET (Ethernet + IPv6 + UDP)

PACKET VALUES		CALCULATIONS
Frame length	**155 bytes**	IPv6 total length = 40 + 101 = 141 bytes
Ethernet header	**14 bytes**	Header overhead = 14 + 40 + 8 = 62 bytes
IPv6 header	**40 bytes**	Payload = 101 - 8 = 93 bytes
UDP header	**8 bytes**	Efficiency = (93 / 155) x 100 = 60%
IPv6 payload length	**101 bytes**	
UDP length / payload	**101 / 93 bytes**	

UDP RESULT: 62 bytes overhead | 93 bytes payload | 60% efficiency

Note: FCS is not included because it is not shown in the captured frame length.

Step-7

Parameter	TCP Packet	UDP Packet
Frame Length	129 bytes	155 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	40 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	115 bytes	141 bytes*
Transport Header Length	20 bytes	8 bytes
Payload Length	75 bytes	93 bytes

Answer these questions

1. What is encapsulation?

Encapsulation is the process of adding protocol headers to data at each network layer before transmitting it over a network. Each layer adds information required for communication.

2. What is the purpose of the Ethernet header?

The Ethernet header is used for communication within a local network. It contains the source MAC address, destination MAC address and EtherType, which identifies the type of data being carried.

3. What information is stored in the IP header?

The IP header contains:

- Source IP address
- Destination IP address
- IP version (IPv4 or IPv6)
- Header length (IPv4)
- Total length (IPv4)
- TTL (IPv4) or Hop Limit (IPv6)
- Protocol information

4. What is the difference between TCP and UDP headers?

TCP Header	**UDP Header**
Minimum 20 bytes	8 bytes
Contains sequence number	No sequence number
Contains acknowledgement number	No acknowledgement number
Contains flags (SYN, ACK, FIN)	No TCP-style flags
Supports reliable communication	Does not guarantee delivery
More header overhead	Less header overhead

5. Why does each protocol add a header?

Each protocol adds a header to provide necessary information for communication, such as addressing, routing, error checking, data delivery and protocol identification.

6. What is packet overhead?

Packet overhead is the extra data added to a packet by protocol headers (and other control information). It does not carry the actual application data but helps in transmitting it.

Example from our TCP calculation:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes

Total overhead = 54 bytes.

7. How does header overhead affect network efficiency?

Header overhead reduces network efficiency because some of the transmitted bytes are used for protocol information instead of actual application data.

Higher overhead means less space is available for application data, while lower overhead generally improves payload efficiency.